# MATH503 S2 2026 Week 4

Monday, 3 August 2026 11:09 am

# MATH503 Lecture

Tuesday
8:00 to 9:50
WA220

## Housekeeping!

- MyiMaths logins
- Attend tutorials
- Check your grades on Canvas for portfolio tasks

# 2.1: Permutations

# Probability

Let $S$ be a sample space of equally likely outcomes, and $E$ an event.

The probability that event $E$ occurs is $P(E) = \dfrac{n(E)}{n(S)}$.

When $n(S)$ is large, it is not possible to list all the outcomes.

Counting principles are needed to find $n(S)$ and $n(E)$.

# The multiplication principle

To find the number of choices, or arrangements of different types of items, we multiply.

To find the total number of arrangements in probability trees we multiply the

| | | | | | | |
|---|---|---|---|---|---|---|
| number of choices at event 1 | X | number of choices at event 2 | X | number of choices at event 3 | | etc |

If we have a choice of A things and then B things and then C things we multiply

If we have choice of A things or B things or C things we add

## Multiplication principle

**Example 1**

A combination lock can be set to open to any 3-letter sequence. How many sequences are possible?

How many sequences are possible if no letter is repeated?

## Multiplication principle

**Example 2**

A computer word is made up of 8 bits, each bit is a digit 0 or 1.

How many different computer words of this type are possible?

# Factorial notation

For any natural number (positive integer) $n$,

$$n! = n \times (n-1) \times (n-2) \cdots \times 3 \times 2 \times 1 \quad \text{and} \quad 0! = 1.$$

**Example 2**

Simplify the following expressions:

- $1! =$
- $5! - 4! =$
- $\dfrac{n!}{(n-2)!} =$
- $\dfrac{(n-2)!}{(n-1)!} =$
- $\dfrac{n!}{0!} =$
- $n(n-1)! =$
- $n! - (n-1)! =$

$2! =$

$\dfrac{6!}{3!} =$

$3! =$

$\dfrac{6!}{2!3!} =$

## Factorials in R

To calculate 5!
<factorial(5)

If the factorial is likely to be very large, we may calculate the natural logarithm of the factorial
< lfactorial(5)

For more help type
< ?factorial

## Permutations

A **permutation** of $r$ elements from a set of $n$ elements is any specific ordering of the $r$ elements.
The number of permutations of $n$ objects taken $r$ at a time is written $P(n, r)$.

**Example 3**:
In a club with 16 members, how many ways can a committee of 3 officers (president, secretary, treasurer) be chosen?
A set of 3 members have to be chosen from 16 members.
The order in which the 3 members are chosen is important.

Use the multiplication principle:

First choice (president):

Second choice (secretary):

Third choice (treasurer):

Number of ways to select the committee:

## Formula for permutations

General rule for $0 \leq r \leq n$: $P(n,r) = \dfrac{n!}{(n-r)!}$

**Proof:**

**Example 4:**
Repeat example 3, but use a permutation to calculate the number of ways in which to select the committee

## In R

A permutation function:

```
perm = function(n, r, log = FALSE){
  if(log == FALSE){
                factorial(n) / factorial(n-r)
  }else if(log == TRUE){
                lfactorial(n) - lfactorial(n-r)
  }}

> perm(16,3)
  3360
> perm(16, 3, log = TRUE)
  8.119696
```

## Combinations or permutations?

*We throw around the term "combination" loosely, and usually in the wrong way. We say things like, "Hey, what's your locker combination?" but what we really ought to be sayings is "Hey, what's your locker* ***permutation****?"*
*Why?*
**Because with permutations we care about the order of the elements, whereas with combinations we don't.**

# 2.2: Combinations

Combinations

Counting techniques

$P(n,r)$ permutations are ordered.

## Combinations

A **combination** is an unordered arrangement of objects or items.

The number of combinations of $r$ objects chosen from $n$ distinct items are written as

$$C(n,r)$$

$$\binom{n}{r}$$

$$^{n}C_{r}$$

$$P(16,3) = \frac{16 \times 15 \times 14 \times 13 \times 12 \times \cdots \times 3 \times 2 \times 1}{13 \times 12 \times 11 \times \cdots \times 3 \times 2 \times 1}$$

## Example 1

How many ways are there to select a committee of 3 people from a group of 16 people?

$$= \frac{16!}{(16-3)!}$$

**Solution**
First calculate the number of ways in which 3 people can be selected from 16 people in an ordered way.
Use a permutation:
$P(16,3) =$

As the positions on the committee are not specified, the order in which the people is selected should not be taken into account.
Each group of 3 will appear a few times in the ordered selections.
How many times?
Use a permutation:
$P(3,3) =$ $3 \times 2 \times 1$

The number of combinations, or unordered selections, of 3 persons from 16 people:

$$C(16,3) = \frac{P(16,3)}{P(3,3)} = \frac{16!}{(16-3)!} \cdot \frac{1}{3!} = \frac{16 \times 15 \times 14}{3 \times 2 \times 1}$$

$$= \frac{3360}{6} = 560$$

## Formula for combinations

In general for $0 \le r \le n$, the number of combinations of $r$ elements chosen from $n$ elements, is given by

$$C(n,r) = \frac{n!}{(n-r)!r!}$$

**Proof:**

number of ways of ~~choosing~~ r objects from n with order is $P(n,r) = \frac{n!}{(n-r)!}$

Number of ways of ordering r objects from a set of r objects is $r!$ so $C(n,r) = \frac{n!}{(n-r)!\,r!}$

## In R

$C(n,r) = \frac{n!}{(n-r)!\,r!}$

To calculate C(16, 3)

```
> choose(16, 3)
  560
```

$r!\,C(n,r) = P(n,r)$

To calculate the log of C(16, 3), useful to avoid numerical errors,

```
> lchoose(16, 3)
  6.327937
```

To get more information, type in R

```
> ? choose
```

## Example 2

| | | |
|---|---|---|
| 3 | 2 | 1 |

A salesman has 10 accounts in a certain city.

(a) In how many ways can he select 3 accounts to call on?

(b) In how many ways can he select 7 accounts to call on?

(c) In how many ways can he select at most 4 accounts to use in preparation of a report?

(d) The salesman has to visit 3 of the accounts picked at random. How many different traveling circuits are possible? $3! = 6$

a) $C(10,3)$

$= \frac{10!}{7!3!} = \frac{10 \times 9 \times 8}{3 \times 2 \times 1}$

$= 120$

b) $C(10,7)$

$= \frac{10!}{3!7!}$

$= 120$

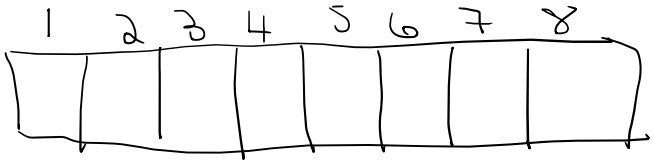
c) $C(10,4) + C(10,3) + C(10,2) + C(10,1) + C(10,0)$

$= \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2} + 120 + \frac{10 \times 9}{2} + 10 + 1$

$= 210 + 120 + 45 + 11 = 386$

## Example 3

How many eight-digit binary numbers have exactly three 1s in them?




$$C(8,3) = \frac{8!}{5!\,3!} = \frac{8\times 7\times \cancel{6}\,\cancel{3}}{\cancel{3}\times\cancel{2}\times 1} = 56$$

## Counting techniques

Four different counting techniques may be used.
In some cases more than one of the techniques may be used to get the same answer:

- multiplication principle,
- permutations,
- combinations,
- tree diagrams.

MATH503 Mathematics for Computing

# 2.3 Counting principles and probability

## Summary of principles of probability

**Definition**
$P(E) = \frac{n(E)}{n(S)}$

**Union rule**
$P(E \cup F) = P(E) + P(F) - P(E \cap F)$

$E$ and $F$ are mutually exclusive if $E \cap F = \emptyset$

$P(E \cup F) = P(E) + P(F)$ if $E$ and $F$ are mutually exclusive

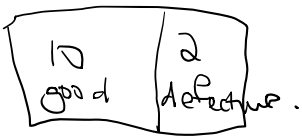
**Complement rule**
$P(E') = 1 - P(E)$

**Product rule**
$P(E \cap F) = P(E|F)P(F)$ or $P(E \cap F) = P(F|E)P(E)$

$E$ and $F$ are independent if $P(E|F) = P(E)$ or $P(F|E) = P(F)$

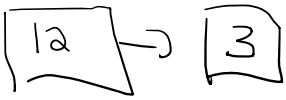
$P(E \cap F) = P(E)P(F)$ if $E$ and $F$ are independent

## Application using combinations




When shipping diesel engines abroad, 12 engines are packed into a container at the plant to be sent to the port.

A spot check is made of the engines after loading.




Three randomly selected engines from a container are tested.

If any of the 3 engines are defective, the container will not be shipped.

Suppose a given container has 2 defective engines.

Find the probability that the container will not be shipped.

Event $E$: container is not shipped

Event $E_1$: one engine fails

Event $E_2$: two engines fail

$E = E_1 \cup E_2$ mutually exclusive so $P(E) = P(E_1) + P(E_2)$

$(1, 2, 3, \ldots 12)$ $n(S) = C(12,3) = \frac{\cancel{12}^{2} \times 11 \times 10}{\cancel{3} \times \cancel{2} \times 1} = 220$

$n(E_1) = C(2,1) \times C(10,2) = 2 \times \frac{10 \times 9}{2} = 90$

$n(E_2) = C(2,2) \times (10,1) = 1 \times 10 = 10$

$P(E_1) = \frac{n(E_1)}{n(S)} = \frac{90}{220}$ $P(E_2) = \frac{n(E_2)}{n(S)} = \frac{10}{220}$

$P(E) = P(E_1) + P(E_2) = \frac{90+10}{220} = \frac{100}{220} = \frac{5}{11}$

$$C(12,3) = \frac{12!}{9!\,3!} = \frac{12\times 11\times 10\times \cancel{9!}}{\cancel{9!}\times 3!}$$

$$= \frac{12\times 11\times 10\times \cancel{(9\times 8\times 7\times 6\times 5\times 4\times 3\times 2\times 1)}}{(\cancel{9!})\times 3!}$$

## Method 1: Union rule

$$\begin{aligned} & P(\text{container is not shipped}) \\ = \; & P(1 \text{ engine fails test}) + P(2 \text{ engines fail test}) \end{aligned}$$

Event $E$: container is not shipped

Event $E_1$: 1 engine fails test

Event $E_2$: 2 engines fail test

Now $E = E_1 \cup E_2$ and

$P(E) = P(E_1) + P(E_2)$ from the union rule, as $E_1 \cap E_2 = \emptyset$.
$E_1$ and $E_2$ are mutually exclusive events.

The sample space $S$ will be all the possible ways in which three engines can be selected from the 12 engines in the container.

## Method 1: Answers

Sample size $S$: all possible ways to select 3 engines from 12 engines

$$n(S) = C(12,3) = \frac{12!}{3!9!} = 220$$

Event $E_1$: 1 fail and 2 successes

$$n(E_1) = C(2,1)C(10,2) = \frac{2!}{1!1!} \times \frac{10!}{2!8!} = 90$$

Event $E_2$: 2 fails and 1 success

$$n(E_2) = C(2,2)C(10,1) = \frac{2!}{2!0!} \times \frac{10!}{1!9!} = 10$$

Find the probability that the container will not be shipped

$$P(E) = P(E_1 \cup E_2) = P(E_1) + P(E_2) = \frac{n(E_1)}{n(S)} + \frac{n(E_2)}{n(S)} = \frac{90}{220} + \frac{10}{220} = \frac{100}{220} = \frac{5}{11}$$

## Method 2: Complement rule

$$
\begin{aligned}
& P(\text{container is not shipped}) \\
= \; & 1 - P(\text{container is shipped}) \\
= \; & 1 - P(0 \text{ engines fail test})
\end{aligned}
$$

Event $E'$: 0 engines fail test

$P(E) = 1 - P(E')$ from complement rule.

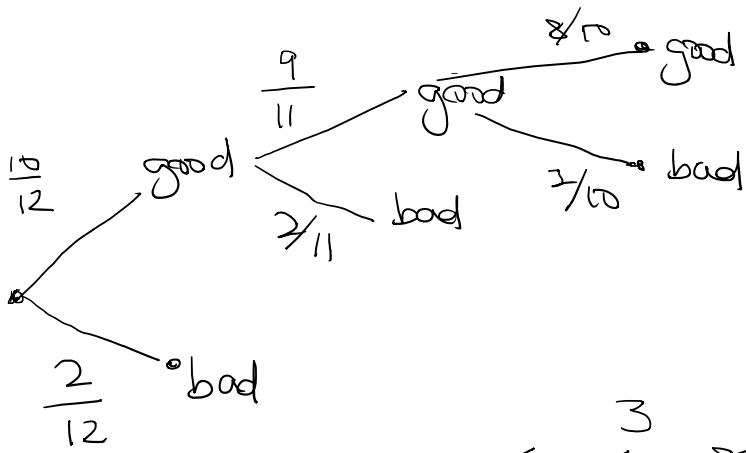
$n(S) = C(12,3) = 220$

$n(E') = C(2,0) \times C(10,3) = \frac{2!}{2!\,0!} \times \frac{10 \times \cancel{9}^{3} \times \cancel{8}^{4}}{\cancel{3} \times \cancel{2}}$

$= 1 \times 120 = 120$

$P(E') = \frac{n(E')}{n(S)} = \frac{120}{220} = \frac{6}{11}$ so $P(E) = 1 - \frac{6}{11}$

$= \frac{5}{11}$




$P(\text{all good}) = P(E') = \frac{\cancel{10}}{\cancel{3}\,\cancel{12}} \times \frac{\cancel{9}^{3}}{11} \times \frac{\cancel{8}^{2}}{\cancel{10}} = \frac{6}{11}$

so $P(E) = 1 - P(E') = 1 - \frac{6}{11} = \frac{5}{11}$

## Method 2: Answers

Sample size $S$: all possible ways to select 3 engines from 12 engines

$$n(S) = C(12,3) = \frac{12!}{3!9!} = 220$$

Event $E$: container is not shipped

Event $E'$: container is shipped

Event $E'$: 0 fails and 3 successes

$$n(E') = C(2,0)C(10,3) = \frac{2!}{0!2!} \times \frac{10!}{3!7!} = 120$$

Find the probability that the container will not be shipped

$$P(E) = 1 - P(E') = 1 - \frac{n(E')}{n(S)} = 1 - \frac{120}{220} = 1 - \frac{6}{11} = \frac{5}{11}$$

## Method 3: Probability tree

Use a probability tree to calculate $P(E') = P(0 \text{ engines fail test})$

## Method 3: Answers

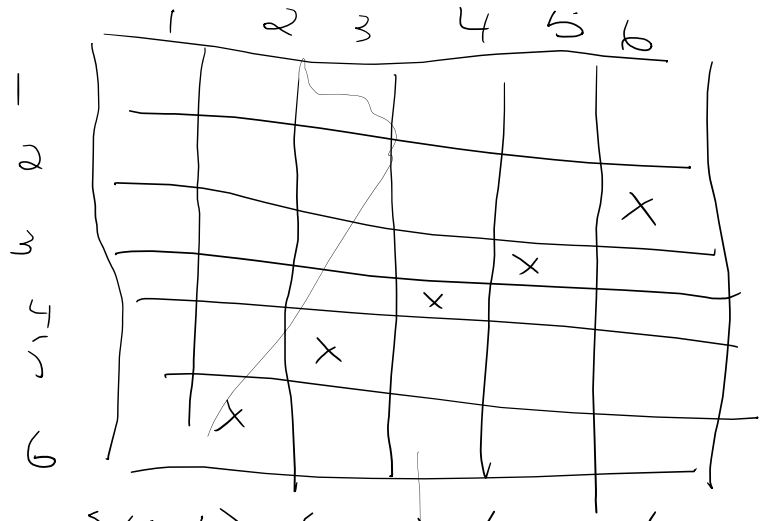
Use a probability tree to calculate $P(E') = P(0 \text{ engines fail test})$

Note that only one path through the probability tree is required:

$$P(E') = P(3\ \textit{successes}, 0\ \textit{fails}) = \frac{10}{12} \times \frac{9}{11} \times \frac{8}{10} = \frac{6}{11}$$

Then use the complement rule:

$$P(E) = 1 - P(E') = 1 - \frac{6}{11} = \frac{5}{11}$$



a) $S = \{(2,6), (3,5), (4,4), (5,3), (6,2)\}$

b) $S = \{(2,6)\ (3,5), (4,4)\}$ order not important.

## What did you get out of today?

What was the purpose of today's lecture?

What is one key thing you learnt?

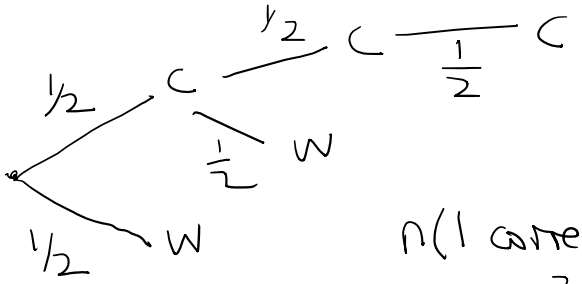
Is there anything that puzzled me?

Student guesses 3 questions T/F

$n(S) = 2 \times 2 \times 2 = 8$

| T/F | T/F | T/F |
|---|---|---|

$P(3 \text{ correct?}) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$




| C | C | C |
|---|---|---|

$n(1 \text{ correct}) = 3$

| C | C | W |
|---|---|---|
| C | W | C |
| W | C | C |

$P(1 \text{ correct answer?}) = \frac{3}{8}$

| | | |
|---|---|---|
| c | w | w |
| w | c | w |
| w | w | c |

$P(2 \text{ correct}) = P(1 \text{ correct})$

$= \frac{3}{8}$

$P(\text{all wrong}) = \frac{1}{8}$

| | | | | | | |
|---|---|---|---|---|---|---|
| 9 | 5 | 5 | 5 | 5 | 5 | 5 |

Not a zero. odd.

$9 \times 5^6$

0, 1, 2, 3, 4, 5, 6, 7, 8, 9

| | | | |
|---|---|---|---|
| 4 | 8 | ? | ? | ? | ? |

$10 \times 10 \times 10 \times 10$

$10^4$

| | | | | | | |
|---|---|---|---|---|---|---|
| 9 | 9 | 8 | 7 | 6 | 5 | 4 |

Not a zero

No repetition.

$9 \times \frac{9!}{3!}$